Here is what all this has to do with the photoelectric effect. Based on the experimental data reviewed above, Einstein suggested incorporating Planck's lumpy picture of wave energy into a new description of light. A light beam, according to Einstein, should actually be thought of as a stream of tiny packets—tiny particles of light—which were ultimately christened photons by the chemist Gilbert Lewis (an idea we made use of in our example of the light clock of Chapter 2). To get a sense of scale, according to this particle view of light, a typical one-hundred-watt bulb emits about a hundred billion billion (1020) photons per second. Einstein used this new conception to suggest a microscopic mechanism underlying the photoelectric effect: An electron is knocked off a metallic surface, he proposed, if it gets hit by a sufficiently energetic photon. And what determines the energy of an individual photon? To explain the experimental data, Einstein followed Planck's lead and proposed that the energy of each photon is proportional to the frequency of the light wave (with the proportionality factor being Planck's constant).

Now, like the children's minimum departure fee, the electrons in a metal must be jostled by a photon possessing a certain minimum energy in order to be kicked off the surface. (As with the children fighting for money, it is extremely unlikely that any one electron gets hit by more than one photon—most don't get hit at all.) But if the impinging light beam's frequency is too low, its individual photons will lack the punch necessary to eject electrons. Just as no children can afford to leave regardless of the huge total number of coins the adults shower upon them, no electrons are jostled free regardless of the huge total energy embodied in the impinging light beam, if its frequency (and thus the energy of its individual photons) is too low.

But just as children are able to leave the warehouse as soon as the monetary denomination showered upon them gets large enough, electrons will be knocked off the surface as soon as the frequency of the light shone on them—its energy denomination—gets high enough. Moreover, just as the dollar-entrusted adult increases the total money thrown down by increasing the number of individual bills tossed, the total intensity of a light beam of a chosen frequency is increased by increasing the number of photons it contains. And just as more dollars result in more children being able to leave, more photons result in